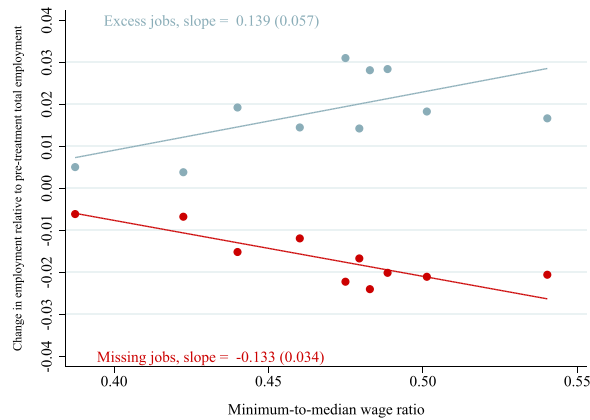


(A) Missing and excess jobs



(B) Employment change

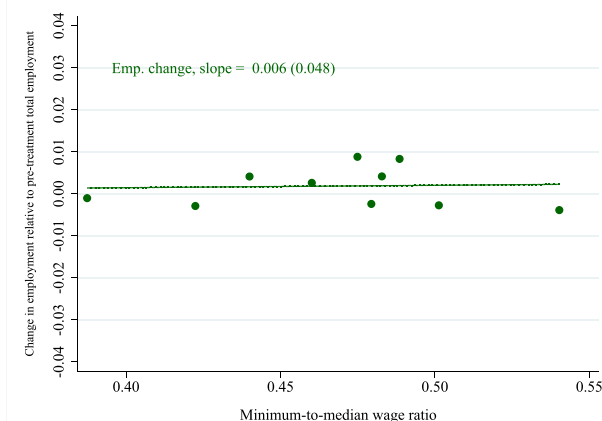


FIGURE V

Relationship between Excess Jobs, Missing Jobs, Employment Change, and the Minimum-to-Median Wage Ratio across Events

The figure shows the binned scatter plots for missing jobs, excess jobs, and total employment changes by the value of the minimum-to-median wage ratio (Kaitz index) for the 130 event-specific estimates. The 130 events exclude 8 minimum-wage-raising events in the District of Columbia, since individual treatment effects are very noisily estimated for those events. (See [Online Appendix](#) Figure A.10 for a raw scatterplot including the events in DC.) The minimum-to-median wage ratio is the new minimum wage MW divided by the median wage at the time of the minimum wage increase (Kaitz index). The bin-scatters and linear fits control for decade dummies, the state-specific unemployment rate at the time of the minimum wage increase, the urban share of the state's population, and an indicator for being a Republican-leaning state. Estimates are weighted by the state populations. The slope (and robust standard error in parentheses) is from the weighted linear fit of the outcome on the minimum-to-median wage ratio.

and the new minimum wage.³² To calculate the former we use the ratio of the minimum wage to the median wage, also known as the Kaitz index (e.g., Lee 1999; Dube 2014; Autor, Manning, and Smith 2016; Manning 2016). When the minimum wage is high relative to the median, it is expected to have a larger bite. Consistent with that expectation, we find that events h with a higher minimum-to-median wage ratio had substantially more missing jobs—the coefficient on $Kaitz_h$ is sizable and statistically significant at -0.133 (std. err. 0.034). At the same time, when we consider excess jobs, we find that the coefficient on $Kaitz_h$ has a very similar magnitude at 0.139 (std. err. 0.057). In other words, when the minimum wage is high relative to the median, the events have a bigger bite and a greater number of missing jobs below the new minimum, but also have a nearly equally sized number of excess jobs at or above the new minimum. As a consequence, the employment effect is virtually unchanged (slope = 0.006, std. err. 0.048) as we consider minimum wages that range between 37% and 59% of the median wage, as shown in Figure V, Panel B. Overall, these findings suggest that the level of the minimum wage increases in the United States that we study have yet to reach a point where the employment effects become sizable.

IV. EMPLOYMENT CHANGES ALONG THE WAGE DISTRIBUTION IN THE CLASSIC TWO-WAY FIXED EFFECT REGRESSION ON LOG MINIMUM WAGE

In the previous section, we estimated the impact of minimum wages on the wage distribution using our event study specification. We found that the effect of the minimum wage was concentrated at the bottom of the wage distribution, and reassuringly we found no indication of considerable employment changes in the upper tail of the wage distribution (see Figure II). The lack of responses \$4 above the minimum wage or higher also implies that the effect of the minimum wage on aggregate employment is close to the estimated employment effect at the bottom of the wage distribution. Such stability of upper-tail employment is consistent with the observation that the share of workers affected by the

32. We control for the state-level unemployment rate at the time of the minimum wage increase, political orientation of the state, urban share of the state, and the decade of the minimum wage increase. However, the results are very similar if we leave out controls; see Online Appendix Figure A.9.

minimum wage changes we study is too small to affect upper-tail employment to a noticeable degree.

In this section, we estimate the effect of the minimum wage on employment throughout the wage distribution using alternative identification strategies to illustrate the advantage of the distributional approach in diagnosing research designs. Recent empirical literature using the classic two-way fixed effect specification with log minimum wage (TWFE-logMW), has found large aggregate disemployment effects in the U.S. context (see [Meer and West 2016](#)).

We decompose the classic two-way fixed effects estimate of log minimum wage on the state-level employment-to-population rate. In [Figure VI](#) we divide the total wage-earning employment in the 1979–2016 Current Population Survey into inflation-adjusted \$1-wage bins by state and by year. Then, for each wage bin, we regress that wage bin's employment per capita on the contemporaneous, four annual lags, and two annual leads of log minimum wage, along with state and time fixed effects.³³ This distributed-lags specification is similar to those used in numerous papers (e.g., [Meer and West 2016](#); [Allegretto et al. 2017](#)).³⁴ The histogram bars show the average post-treatment effect divided by the sample average employment-to-population rate,³⁵ while the dashed purple line (color version available online) plots the running sum of the employment effects of the minimum wage up to the particular wage bin. The final purple bar represents the estimated effect on the aggregate employment to population rate.

[Figure VI](#) shows that on average, minimum wage shocks are associated with large employment changes in the real dollar bins in the \$6 to \$9/hour range. There is a sharp decrease in

33. In the TWFE-logMW model, the point estimates for the leads and lags show the impact relative to the employment in the third year or earlier. Once we normalize the TWFE-logMW estimates to the first lead, we can report three leads and four lags, similarly to our benchmark estimates.

34. [Meer and West \(2016\)](#) present unweighted results on the total employment effect of the minimum wage. Here we present estimates weighted by the population size because it is more standard in the literature and also closer to our event study estimates. However, as we show in [Online Appendix Figure A.11](#), the unweighted estimates are similar.

35. We construct the cumulative response over event dates 0, 1, ..., 4 relative to event date -1 by successively summing the coefficients for contemporaneous and lagged minimum wages. We then average the cumulative responses over dates 0, 1, ..., 4. This average post-treatment effect is analogous to what we did in our event-based analysis in the previous sections.

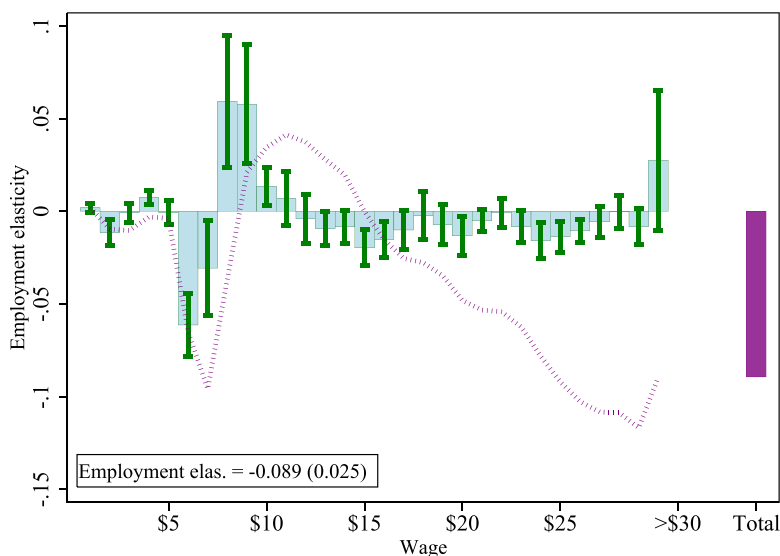


FIGURE VI

Impact on Employment throughout the Wage Distribution in the Two-Way Fixed Effects Model on Log Minimum Wages

The figure shows the effect of the minimum wage on the wage distribution in fixed effects (TWFE-logMW) specification. We estimate two-way (state and year) fixed effects regressions on the contemporaneous log minimum wage, as well as on four annual lags and two annual leads. For each wage bin we run a separate regression, where the outcome is the number of jobs per capita in that state-wage bin. The cumulative response for each event date 0, 1, ..., 4 is formed by successively adding the coefficients for the contemporaneous and lagged log minimum wages. The green histogram bars (color version available online) show the mean of these cumulative responses for event dates 0, 1, ..., 4, divided by the sample average employment-to-population rate—and represent the average elasticity of employment in each wage bin with respect to the minimum wage in the post-treatment period. The 95% confidence intervals around the point estimates are calculated using clustered standard errors at the state level. The dashed purple line plots the running sum of the employment effects of the minimum wage up until the particular wage bin. The rightmost solid purple bar is the elasticity of the overall state employment-to-population with respect to the minimum wage, obtained from regressions where the outcome variable is the state-level employment-to-population rate. In the bottom left corner we also report the point estimate on this elasticity with standard errors that are clustered at the state level. Regressions are weighted by state population. The figure highlights that large aggregate dis-employment effects are often driven by shifts in employment at the upper tail of the wage distribution.

employment in the \$6/hour and \$7/hour bins, likely representing a reduction in jobs paying below new minimum wages, and a sharp rise in the number of jobs in the \$8/hour and \$9/hour wage bins, likely representing jobs paying above the new minimum. At